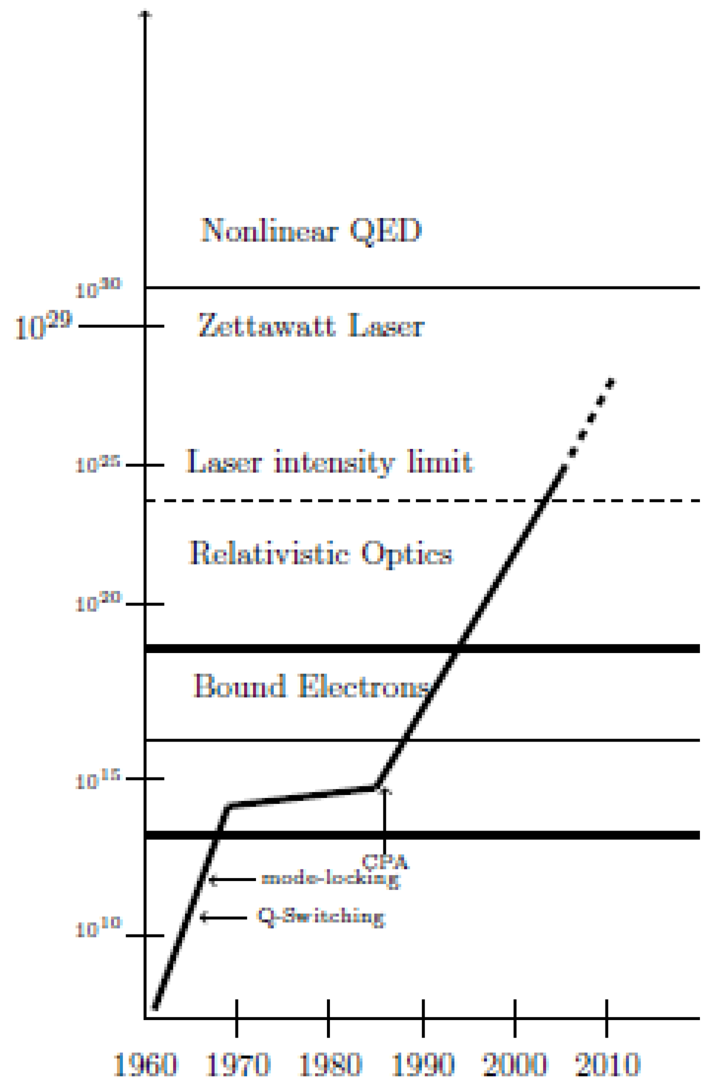
Atomic, Molecular and Optical (AMO) Physics

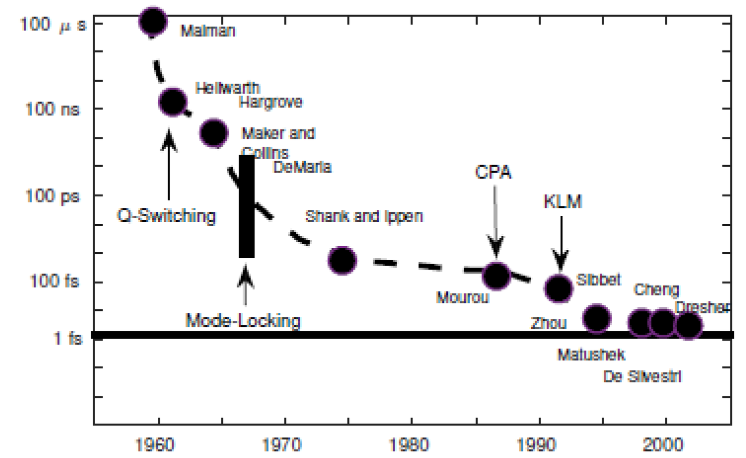
Controling matter through the help of radiation

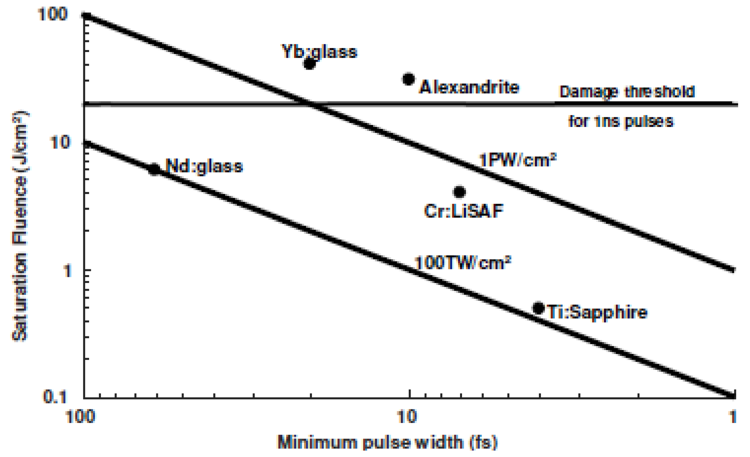
High harmonic generation (HHG) and photoelectron momentum distribution (PEMD) are some of the most important processes driving the study of matter-radiation interaction and will lead to important advances in fundamental science and engineering. Through HHG, for example, it is possible to produce coherent radiations outperforming the external fields in regards to frequency and pulse duration. The current theory, valid for stationary systems, falls short at the description of nonstationary systems, since it ignores radiations induced by the external force. Following Yangaliev *et al.*[1], we aim at extending the quantum theory of radiation for real nonstationary systems, i.e in 3 dimensions and for realistic potentials.

1. Background

Evolution of the laser field

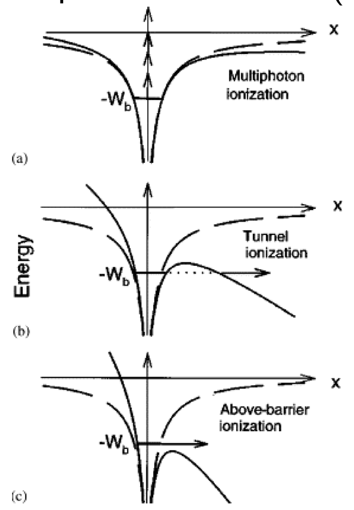




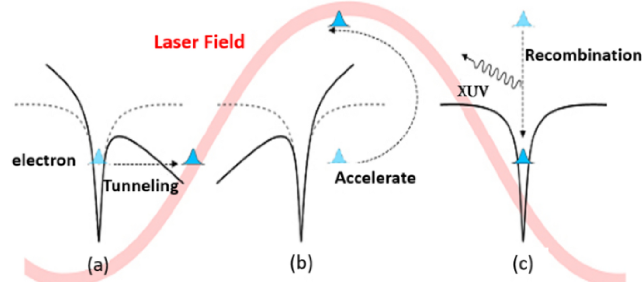


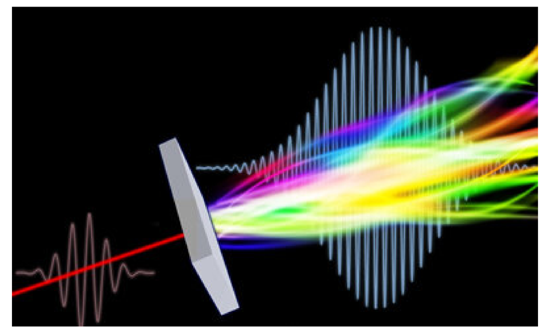
Basic strong field phenomena

1. Multiphoton Ionization (MPI)

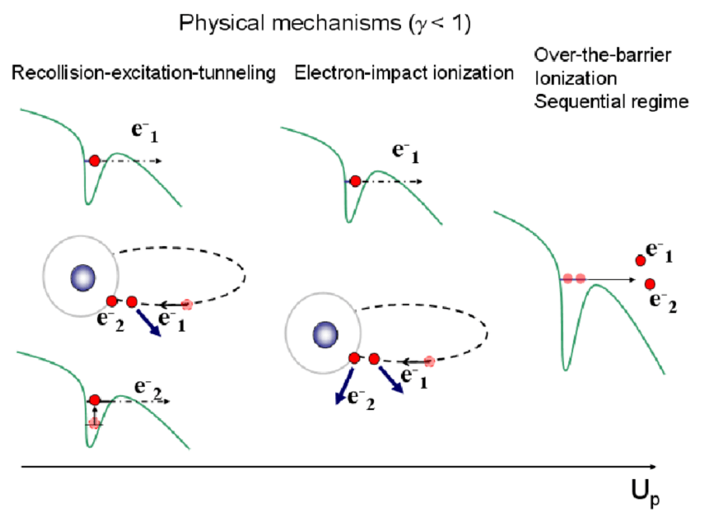


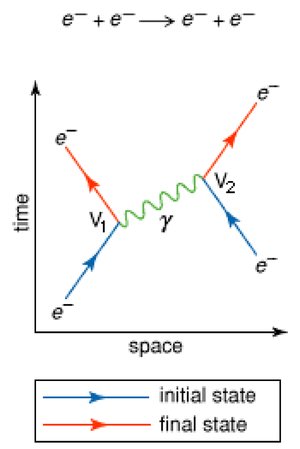
2. High harmonic generation (HHG)





3. Nonsequential double ionization (NSDI)





2. Theory

Strong field laser interaction with atoms
Schrödinger equation

$$i\frac{\partial\,|\Psi(t)\rangle}{\partial t} = \left[\hat{H}_0 + \boldsymbol{F}(\boldsymbol{t})\cdot\hat{\mathbf{r}}\right]|\Psi(t)\rangle\,,$$

$$|\Psi(t)\rangle = \sum_n A_n(t)e^{-iE_nt}\,|\psi_n\rangle + \int B_k(t)e^{-iE_kt}\left|\psi_k^{(-)}\right\rangle dk.$$

$$\boldsymbol{F}(t) = F_0 f(t)\exp(\omega t+\varphi),\qquad p(t) = -\langle\psi(\boldsymbol{r},t)|\,\hat{\boldsymbol{p}}\,|\psi(\boldsymbol{r},t)\rangle\,,\qquad f(\omega) = \int e^{i\omega t}f(t)dt \quad \text{Fourier transform}$$

Quantities relevant to the dynamics of the system are evaluated by exploiting the wavepacket $|\Psi(\boldsymbol{r},t)\rangle$. The probability to find the system in any of the bound states $|\psi_n\rangle$ or excited states $\left|\psi_k^-\right\rangle$ or excited states

$$P_n = \left|\left\langle\psi_n^-|\Psi(t)\right\rangle\right|^2,$$

The survival probability and the ionization probabilities are

$$P_{\rm surv} = \sum_n \left|\langle\psi_n|\Psi(t)\rangle\right|^2,\quad E_n \leqslant 0,\qquad P_{\rm ion} = 1 - P_{\rm surv}.$$

and the probability density of the wavepacket is

$$\mathcal{D}(\boldsymbol{r},t) = |\Psi(\boldsymbol{r},t)|^2 = \left|\sum_n A_n(t)e^{-iE_nt}\,|\psi_n\rangle + \int B_k(t)e^{-iE_kt}\left|\psi_k^{(-)}\right\rangle dk.\right|^2$$